# Didactic Manual

**big.bi → bi.gbi**

**Articulatory Simulation of the Verbal Transformation**

Reproduction of the demonstrations described in
*"Why can big.bi be changed to bi.gbi?*
*A mathematical model of syllabification and articulatory synthesis"*
arXiv:2307.02299 (Berthommier, July 2023)

| | |
|---|---|
| **Model:** | VLAM (Maeda, 7 parameters) |
| **Code:** | Python 3.9+ (MIT License) |
| **Dependencies:** | numpy, scipy, matplotlib, opencv-python |
| **Repository:** | `bigbi-article-repo` |

*This manual provides step-by-step instructions for reproducing the verbal transformation simulations, including the polar trajectory visualization, the U-shaped Tp variation, and the comparison with the article's Figure 4.*

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# Contents

**1 Introduction** **3**
1.1 What this repository provides . . . 3
1.2 The VLAM model . . . 3
1.3 Article parameters . . . 4

**2 The Mathematical Model** **5**
2.1 Polar coordinates and targets . . . 5
2.2 Arc trajectories . . . 5
2.3 Anchoring vowels: Vo and Ve . . . 5
2.4 Superposition . . . 5

**3 The Syllable Graph Architecture** **6**
3.1 CV graph . . . 6
3.2 CVC graph . . . 6
3.3 CCV graph (clusters) . . . 6
3.4 The fusion mechanism . . . 6

**4 The $\delta_o$ and $\delta_e$ Parameters** **7**
4.1 Implementation . . . 7
4.2 Effect on the anchoring vowels . . . 7
4.3 Word-end decay: no held @ . . . 7

**5 Reproduction Guide** **8**
5.1 Installation . . . 8
5.2 Demonstration 1: big.bi → bi.gbi (Article §4) . . . 8
5.3 Demonstration 2: U-shaped $T_p$ variation . . . 8
5.4 Demonstration 3: Comparison with the article . . . 9

**6 Polar Trajectory Reconstruction** **9**
6.1 Reconstruction rules . . . 9
6.2 Display curvature and $\nu$ convention . . . 10
6.3 Comparison with the article's Figure 4 . . . 10

**7 Results and Duration Analysis** **10**
7.1 Duration reduction . . . 10
7.2 Audio verification: per-segment WAVs, /i/ height, valrect . . . 11
7.3 Perceptual salience of the schwa @ . . . 11

**8 Code Structure** **12**

8.1 Modified files (vs original timit-to-Maeda) . . . . . 12

8.2 Simulation scripts . . . . . 12

**9 Quick Demo** **12**

**10 Article Demonstrations** **13**

10.1 Demo 1: /ibia/ VCV synthesis (Figure 1) . . . . . 13

10.2 Demo 2: Locus equations (Figure 3) . . . . . 13

10.2.1 Measurement-protocol sensitivity . . . . . 15

10.3 Demo 3: Trough effect (Figure 1b) . . . . . 15

10.4 Demo 4: Six consonant cluster pairs . . . . . 17

10.5 Figure 4: big.bi | bi.gbi (article format) . . . . . 18

**11 Interactive HTML Simulations** **18**

11.1 big.bi → bi.gbi (T=16, starts at COEFCEN=0.5) . . . . . 18

11.2 ib ib → bi bi (NON-REVERSIBLE U-shape) . . . . . 19

11.3 ib ib → bi bi (U-shape, reversible) . . . . . 19

**12 Conclusion** **19**

# 1 Introduction

This manual documents the Python implementation of the articulatory speech synthesis model described in arXiv:2307.02299 [1]. The model, based on VLAM (a Maeda articulatory synthesizer with 7 parameters), simulates the **verbal transformation** from `big.bi` to `bi.gbi` — a phenomenon where the syllable boundary shifts as the pause between syllables is shortened and the anchoring vowels (Vo, Ve) approach the main vowel V.

The article's central claim (§4) is:

> *"When $\delta_o = \delta_e$ tends to 1 the graph reorganization can occur because two successive superimposed segments of duration $2 \times 2T$ anchored to the same vowel $V_o = V_e = V$ can fuse into a single one of duration $3T$."*

This repository provides the code to reproduce this transformation step by step, with audio synthesis, polar trajectory visualization, and accessible video output.

## 1.1 What this repository provides

1. The `synthSYL` package (modified from FBerthommier/timit-to-Maeda) with the $\delta_o$ and $\delta_e$ parameters added to `panphon_pipeline()`.
2. The VLAM synthesizer (`vlam.py`) for audio generation from Maeda parameters.
3. Simulation scripts for three demonstrations:
   - `big.bi` → `bi.gbi` (the article's main result, §4)
   - `ib ib` → `bi bi` (the classical transformation, §3)
   - U-shaped $T_p$ variation for perceptual demonstration
4. Polar trajectory reconstruction (two-branch: vocalic $z_v$ and consonantal $z_c$), using primitives from FBerthommier/covtl-pipeline.
5. Dual-panel video generation (sagittal vocal tract | polar plot) with synchronized audio and phoneme labels.

## 1.2 The VLAM model

VLAM (Vocal tract Learner Articulatory Model) is the Maeda model with 7 articulatory parameters:

| Index | Symbol | Description |
|---|---|---|
| 0 | J | Jaw opening |
| 1 | B | Tongue body position |
| 2 | D | Tongue dorsum height |
| 3 | T | Tongue tip position |
| 4 | LP | Lip protrusion/rounding |
| 5 | LH | Vertical lip opening |
| 6 | Hy | Larynx height |

**Indexing convention.** Two numbering conventions coexist and must not be mixed:

- the **article** numbers the parameters 1–7 (1=Jaw, 2=Body, 3=Dorsum, 4=Tip, 5=LipP, 6=LipH, 7=Hy); its selection vectors $S_c$ and $S_v$ (e.g. $S_c = \{1, 2, 6\}$) use this 1-based numbering;
- the **code** (and the table above) numbers them 0–6; the registry selection vectors are 0-based, e.g. /b/ has `params=[0,1,5]`.

Correspondence: article $\{1, 2, 6\}$ = code $\{0, 1, 5\}$ = {Jaw, Body, LipH}; article $\{1, 2, 3, 4\}$ = code $\{0, 1, 2, 3\}$ = {Jaw, Body, Dorsum, Tip}; article $S_v = \{3, 4, 5, 7\}$ = code $\{2, 3, 4, 6\}$ = {Dorsum, Tip, LipP, Hy}. Throughout this manual, $S_c/S_v$ sets are quoted in the article's 1-based numbering (as in the article), while parameter indices mentioned in the text refer to the 0-based code indices unless stated otherwise.

The coordination function (article Eq. 1) maps a polar target $(\rho, \theta)$ to the 7 Maeda parameters:

$$P_i = \Omega_i + \rho \cdot \Psi_{1,i} \cdot \cos(\Psi_{2,i} - \theta) \tag{1}$$

where $\Omega$, $\Psi_1$, and $\Psi_2$ are calibration constants (article Table 1). The CO matrix matches the article's Table 1 except for the Tip row, where two parameters differ: $\Omega = 0.5$ (article: 0) and $\Psi_1 = -2.75$ (article: $-3.0$) in `synthSYL/constants.py`; all other rows are identical. The deviation is inherited from the original timit-to-Maeda registry and is quantified in the Differential Supplement, §4 (up to 0.70 Maeda units on Tip for /d/, /g/; zero for /b/ and /gb/).

## 1.3 Article parameters

The article (§2.1) specifies the following arc parameters:

- $K_{\text{voy}} = 30$ — vowel arc curvature
- $K = 10$ — consonant arc curvature
- $P_{\text{exp}} = 1$ — velocity profile exponent ($\rho(t) = \cos(\theta(t)/2)$, not $\cos^2$)
- $T = 16$ — time step (didactic mode)
- $\nu = -1$ — coarticulation index

These are set in `synthSYL/constants.py`:

```
DEFAULT_T: int = 16
DEFAULT_K: int = 10          # consonant arc curvature
DEFAULT_KVOY: int = 30       # vowel arc curvature
DEFAULT_PEXP: int = 1        # velocity profile exponent
DEFAULT_NU: int = -1
```

The article notes: *"When $K$ is large ($K = 30$ for vowel arcs and $K = 10$ for consonant arcs), the trajectories become straight lines."* With $P_{\text{exp}} = 1$, the velocity profile follows $\cos(\theta/2)$ (a cosine half-bell) rather than $\cos^2(\theta/2)$ (a Hanning window).

**Note on display curvature:** The polar trajectory visualization uses $K = 30$ for both vocalic and consonantal branches (the covtl-pipeline reference display curvature), not the engine's $K = 10$ / $K_{\text{voy}} = 30$. This is a deliberate visualization choice that produces moderate, visible curvature for both branches.

# 2 The Mathematical Model

## 2.1 Polar coordinates and targets

Each phoneme target is a point $(\rho, \theta)$ in the complex plane:

- **Vowels**: $\rho \in [0, 1]$, $\theta \in [0, 2\pi)$
- **Consonants**: $\rho \in [0, 1.2]$, $\theta \in [0, 2\pi)$

For example, the vowel /i/ is at $(\rho = 0.9, \theta = 5\pi/3)$ and the consonant /b/ is at $(\rho = 1.2, \theta = \pi/3)$ (registry values; the article publishes $\rho = 1$ for the corner vowels and for /b/ — see the Differential Supplement, §4, row 3).

## 2.2 Arc trajectories

Motion between two targets $(\rho_1, \theta_1)$ and $(\rho_2, \theta_2)$ is produced by the arc function (article Eq. 2):

$$z(t) = (1 - r_k) \cdot \rho_1 e^{i\theta_1} + r_k \cdot \rho_2 \cdot e^{i(\theta_2 + \frac{\nu}{K} \cdot \text{th})} \tag{2}$$

where $r_k = \cos^{P_{\text{exp}}}(\text{th}/2)$ is the velocity profile, $\text{th} \in [0, \pi]$ is a half-turn sweep, $\nu = \pm 1$ controls the curvature direction, and $K$ is the curvature constant ($K = 30$ for display, $K = 10$ for consonant arcs, $K_{\text{voy}} = 30$ for vowel arcs in the synthesis engine).

**Important:** The article uses $P_{\text{exp}} = 1$, so $r_k = \cos(\text{th}/2)$ (a cosine half-bell, *without* the power of 2). This is the form used for all demonstrations in this repository. The original `synthSYL` package used $P_{\text{exp}} = 2$ ($r_k = \cos^2(\text{th}/2)$, a Hanning window), which has been corrected to match the article.

## 2.3 Anchoring vowels: Vo and Ve

The article (§2.2) introduces two anchoring vowels:

$$V_o = (\delta_o \cdot \rho_V, \theta_V) \tag{3}$$
$$V_e = (\delta_e \cdot \rho_V, \theta_V) \tag{4}$$

- $V_o$ (onset): anticipation of the vowel before the consonant. The value of $\delta_o$ determines the degree of anticipation.
- $V_e$ (end): centralization of the coda vowel (schwa-like vocoid). Modeled with $\delta_e = 0.5$ (salient) or $\delta_e = 0.7$ (audible).

When $\delta_o = \delta_e \to 1$, both $V_o$ and $V_e$ approach $V$ (the full vowel), enabling the fusion of adjacent syllable segments.

## 2.4 Superposition

The final parameter frame is a blend (article Eq. 4):

$$P(t) - \Omega = \text{Re}\left[S_v \cdot \Psi \cdot \bar{z}_v(t) + S_c \cdot \Psi \cdot \bar{z}_c(t)\right] \tag{art. 4}$$

where $S_v$ and $S_c$ are selection vectors (binary masks over the 7 parameters), $z_v(t)$ is the vocalic trajectory, and $z_c(t)$ is the consonantal trajectory.

# 3 The Syllable Graph Architecture

## 3.1 CV graph

A CV syllable (e.g., /bi/) has 3 nodes and 4 arcs:

- Nodes: $V = \{V_o, C, V\}$
- Locations: $L = \{(\delta_o \rho_V, \theta_V),\ (\rho_C, \theta_C),\ (\rho_V, \theta_V)\}$
- Arcs: $E = \{(V_o, C), (C, V), (V_e, V), (V, V)\}$
- Durations: $T_p = \{T_1, T_2, 2T_2, T\}$

## 3.2 CVC graph

A CVC syllable (e.g., /big/) has 5 nodes, symmetric with respect to the CV graph:

- Nodes: $V = \{V_o, C_1, V, C_2, V_e\}$
- Locations: $L = \{(\delta_o \rho_V, \theta_V),\ (\rho_{C_1}, \theta_{C_1}),\ (\rho_V, \theta_V),\ (\rho_{C_2}, \theta_{C_2}),\ (\delta_e \rho_V, \theta_V)\}$

## 3.3 CCV graph (clusters)

A CCV syllable (e.g., /gbi/) has 4 nodes and 5 arcs:

- Nodes: $V = \{V_o, C_1, C_2, V\}$
- Arcs: $E = \{(V_o, C_1), (C_1, C_2), (C_2, V), (V_e, V), (V, V)\}$
- The consonantal branch visits $C_1$ then $C_2$ in sequence.

## 3.4 The fusion mechanism

The article (§4) describes the fusion:

> *"When $\delta_o = \delta_e$ tends to 1 the graph reorganization can occur because two successive superimposed segments of duration $2 \times 2T$ anchored to the same vowel $V_o = V_e = V$ can fuse into a single one of duration $3T$."*

The gain is twofold:

1. **Consonant clustering**: reduction of one period $T$ (from 2 separate consonant gestures to 1 fused cluster gesture).
2. **Node loss**: the $V$ node between the two syllables disappears (since $V_o = V_e = V$), reducing the graph complexity.

Additionally, the selection vector switches (article's 1-based numbering; code indices subtract 1):

- Before fusion: $S_c = \{1, 2, 3, 4\}$ for /g/ and $S_c = \{1, 2, 6\}$ for /b/ (two separate gestures)
- After fusion: $S_c = \{1, 2, 3, 6\}$ for /gb/ (one fused gesture)

# 4 The $\delta_o$ and $\delta_e$ Parameters

## 4.1 Implementation

The $\delta_o$ and $\delta_e$ parameters are added to the `panphon_pipeline()` function as optional keyword arguments:

```
from synthSYL import panphon_pipeline

result = panphon_pipeline(
    "big bi",
    T=16,
    short_pause_duration_factor=1.0,
    delta_o=0.5,    # Vo = (0.5 * rho_V, theta_V)
    delta_e=0.5,    # Ve = (0.5 * rho_V, theta_V)
)
```

The default values are $\delta_o = \delta_e =$ `COEFCEN` $= 0.5$, preserving backward compatibility with the original package.

## 4.2 Effect on the anchoring vowels

Table 1 shows the effect of $\delta$ on the anchoring vowels for the vowel /i/ ($\rho_V = 0.9$, $\theta_V = 5\pi/3$):

Table 1: Effect of $\delta$ on $V_o$ and $V_e$ (for /i/)

| $\delta$ | $V_o = V_e$ $\rho$ | Perception | Article term |
|---|---|---|---|
| 0.50 | 0.450 | Salient schwa @ | “salient” |
| 0.70 | 0.630 | Audible schwa | “audible” |
| 0.85 | 0.765 | Residual schwa | — |
| 0.95 | 0.855 | $V_o \approx V_e \approx V$ | — |
| 1.00 | 0.900 | $V_o = V_e = V$ (fusion) | — |

## 4.3 Word-end decay: no held @

A reduced word-end vowel is not pronounced as a tensed vowel — one does not pronounce “ibe”. The engine therefore behaves like the reference `timit-to-Maeda`: the trajectory *reaches* the $V_e$ anchor and decays immediately, with no extra hold. The reference `_handle_b_terminal()` chains the decay directly:

```
elif A.kind == "V":
    if B.kind == "pause":
        pt_A = _safe_pt(A)
        _append_decay_block(blocks, block_info, pt_A, T_cons, pre_amp=1.0)
```

This repository matches the reference: the final vowel of a CV word is still held through the regular last-anchor hold, and there is no extra pre-decay $V_e$ plateau. /ibia/ is unaffected (90 steps), while /ib/ is 112 steps = 1120 ms (arc + attack + plateau + cluster + decay + arc).

Note on the visible timing: the decay block is itself a T-step stationary hold at the $V_e$ anchor with the amplitude envelope ramping down, and the terminal arc departs from $V_e$, so the trajectory is SEEN sitting at (or near) $V_e$ for $\approx 2T$ (e.g. /ib/: arrival at 800 ms, envelope at zero by ≈910 ms, neutral reached at 1120 ms). This lengthening is EXAGGERATED relative to

natural speech, but it is exactly what makes the $V_e$ anchoring and the schwa-to-vowel gradation visible and audible in the demonstrations.

# 5 Reproduction Guide

## 5.1 Installation

```
# Clone the repository
git clone https://github.com/<your-org>/bigbi-article-repo.git
cd bigbi-article-repo

# Install Python dependencies
pip install -r requirements.txt

# (Optional) Install ffmpeg for video generation
sudo apt install ffmpeg  # Linux
# brew install ffmpeg     # macOS
```

## 5.2 Demonstration 1: big.bi → bi.gbi (Article §4)

This is the main demonstration, showing the verbal transformation with $T = 16$ (didactic mode), starting at $\delta = 0.50$ for a salient boundary schwa @. **Dot form**: big.bi is synthesized as the dotted word `big.bi` — the ". " between /g/ and /b/ is a C.C syllable boundary with *no pause* (the reference Timit-to-Maeda semantics: the Ve of "big" is coarticulated with the /i/ of "bi", the boundary anchor weighted by COEFCEN). There is therefore *no* $T_p$/pause dimension in this sweep: $\delta$ alone drives the transformation (boundary anchor $= \delta \cdot \rho_i$; at $\delta = 1$ the anchor is the full /i/, fusion-ready).

```
# Run the sweep (14 segments: 11 "big.bi" + 3 "bi.gbi")
python scripts/run_bigbi_polar_sweep.py

# Generate the dual-panel video (sagittal | polar)
python scripts/make_bigbi_dual_video.py

# Generate the self-contained HTML with embedded video
python scripts/embed_video_in_html.py en
```

The output is in `output/bigbi_polar_sweep/`:

- `dual/dual_progressive.mp4` — dual-panel video (≈ 24.2 s, 1.4 MB)
- `polar/polar_progressive.mp4` — polar-only video (3.3 MB)
- `index_embedded.html` — self-contained HTML (re-embedded in `simulations/`)

## 5.3 Demonstration 2: U-shaped $T_p$ variation

The U-shaped variation lets the listener clearly perceive: `ib ib` at the start → `bi bi` at the bottom (fusion) → `ib ib` again at the end. This $T_p$ sweep is the /ibi/ supplement demonstration (§3 classical transformation), where the inter-word pause IS the object of study; it is distinct from the big.bi sweep of Demonstration 1, which — in the dot form — carries no $T_p$ dimension.

```
python scripts/run_ibbi_ushape_sweep.py
```

The sweep is a **continuous U-shaped ramp** of **10 segments**: $T_p$ decreases to 0 with $\delta$ correlated at every point ($\delta = 0.5 + 0.5\,(1 - T_p/160)$), then increases back to the 160-ms baseline; consecutive segments flow directly into one another (no restarts):

| | Segments | $T_p$ (ms) | $\delta$ |
|---|---|---|---|
| descent | 1–4 | 160, 120, 80, 40 | 0.50 . . . 0.875 |
| bottom (fusion, `bi bi`) | 5 | 0 | **1.000** |
| ascent | 6–9 | 40, 80, 120, 160 | 0.875 . . . 0.50 |
| end baseline | 10 | 160 | 0.500 |

Duration ≈ 18.6 s. Note: the coda @ is *not* held — the trajectory reaches the word-end anchor and decays immediately (no "ibe").

### 5.4 Demonstration 3: Comparison with the article

To compare our polar figures with the article's Figure 4:

```
# Generate the comparison figure
python scripts/compare_article_polar.py
```

This produces a 2×2 grid: article Figure 4 (top) vs our simulation (bottom), for both `big.bi` ($\delta = 1$) and `bi.gbi`.

## 6 Polar Trajectory Reconstruction

The two-branch polar visualization (vocalic $z_v$ in red, consonantal $z_c$ in blue) is rebuilt from the engine's **recorded block sequence** by `scripts/polar_sync.py` (`record_pipeline` instruments the trajectory helpers without touching the engine files; `build_branches` then draws both branches from the blocks the engine actually walked). The original `Syllable_Synthesis` display anchoring is used:

- every sub-arc carries the phase on the **arrival-side** point ($p_a$) with the departure-side point $p_d$ held fixed ($p_d$ = the consonant on $z_c$ approach legs, $V_1$ on vocalic backgrounds) — this is the reference convention, not an option;
- the approach leg sweeps $\theta \in [0, \pi]$ (opint = 0), the release legs and vocalic backgrounds $\theta \in [-\pi, 0]$ (opint = −1, first sample dropped, for the releases);
- $\rho = \cos(\theta/2)$ ($P_{\text{exp}} = 1$).

### 6.1 Reconstruction rules

| **Block kind** | **Branch behavior** |
|---|---|
| plateau, attack, decay | red: stationary anchor; blue: inactive |
| background, pause | red: arcplot arc ($V_1 \to V_2$) with $K = 30$, $\nu_v = -1$; blue: inactive |
| cluster ($n = (m{+}1)T$ steps) | red: arcplot arc ($V_1 \to V_2$) over the whole block; blue: sub-arcs $V_1 \to C_1 \to \cdots \to C_m \to V_e$, $T$ steps each, $K = 10$, $\nu_c = +1$, approach leg $[0, \pi]$ then release legs $[-\pi, 0]$ — the closed V→C→V gesture is a **teardrop** (belly at the vowel, point at the consonant), as in the article's Figure 4 |

### 6.2 Display curvature and $\nu$ convention

The display uses the article values $K = 30$ for $z_v$ and $K = 10$ for $z_c$ (§2.1), and the per-branch display convention $\nu_v = -1$, $\nu_c = +1$ (the article's planning-figure convention; the engine itself drives every arc with the single value $\nu = -1$). The legacy `polar_arc` form for $z_v$ (mirrored phase sweep) remains available as `zv_form="polar_arc"` for the record.

### 6.3 Comparison with the article's Figure 4

Under the article figure conditions (COEFCEN = 1, VOYDEB = 0.5, $T = 16$), the reproduced planning panels (under `docs/figures/`) are:

- `article_format/figure4_bigbi_bigbi.png` (article-format figure, §11.6);
- `fig_bigbi_article_conditions_nu_inv.png` (big.bi | bi.gbi, two panels);
- `fig_words_article_conditions_nu_inv.png` (per-word panels: "big", "bi", ".gbi").

They show:

- big.bi: **3** separate $z_c$ gestures — /b/ (onset), palatal /g/ (coda), /b/ (word-2 onset, a closed loop anchored on the previous $V_e$);
- bi.gbi: **2** gestures — /b/ + the fused /gb/ cluster (single span visiting /g/ then /b/);
- the red $z_v$ branch: schwa loop at the word onset ($V_o$ at half radius), full vowel elsewhere ($V_e = V$ under COEFCEN = 1).

The article's Figure 4 uses a different polar target table (the AP-model layout), so the panel-level comparison is gestural (same teardrop topology, same gesture counts), not a point-wise overlay.

## 7 Results and Duration Analysis

### 7.1 Duration reduction

The article predicts a duration reduction when transitioning from `big.bi` ($\delta < 1$) to `bi.gbi` ($\delta = 1$):

> *"At first, the gain of consonant clustering is a reduction of one period $T$. The graph complexity is corollary reduced with a decay of the number of nodes with the loss of $V$."*

With $T = 16$, in the dot form (no pause — the block layout does not depend on $\delta$, only the anchor values do), our simulation shows:

| **Configuration** | $\delta$ | Steps | Duration (ms) | Reduction |
|---|---|---|---|---|
| big.bi (dot, start) | 0.50 | 176 | 1760 | — |
| big.bi (dot, pre-fusion) | 0.995 | 176 | 1760 | — |
| bi.gbi (fused) | 1.00 | 160 | 1600 | −160 ms ($-1T$) |

The observable fusion gain is $1T = 160$ ms, from consonant clustering: the fused /gb/ cluster chains the /g/→/b/ sub-arcs directly through the shared vowel anchor, and the separate syllable-2 onset gesture disappears. This matches the article's "the gain of consonant clustering is a reduction of one period $T$": in the dot form the '.' is a C.C syllable boundary without pause, and the only latencies are the stop closures.

## 7.2 Audio verification: per-segment WAVs, /i/ height, valrect

Every simulation writes its segments as SEPARATE WAV files, so the /i/ vowels can be compared by ear:

- `output/ibbi_ushape_en/wav/synth_sNN_tpXXX.wav` — reversible U-shape, one file per segment;
- `output/ibbi_ushape_nonrev_en/wav/` — non-reversible;
- `output/bigbi_polar_sweep/wav/synth_dXXX.wav` — big.bi → bi.gbi sweep;
- `output/article_demos/` and `output/trough_ablations/` — article reproductions and trough-effect measurements.

Measured on the current 10-segment sweeps (autocorrelation F0 on the vowel plateaus): the median F0 is constant across segments (≈ 119–124 Hz, cf0 = 1.0 everywhere) — the /i/ vowels all sit at the same pitch; the small median shift between "ib ib" and "bi bi" segments is a formant-related measurement bias, not a pitch difference. No digital clipping anywhere (<0.02 % of samples at peak). `valrect` (the soft-rectification smoothness of the vocal-tract area function, `soft_rect(x, s)` with $s = 0 \equiv$ hard rectification) is **1.10 in every script** — a documented difference: the article publishes no such setting, and 0.75 is the historical timit-to-Maeda default; the raised value reduces the perceived fizz on /i/. Measured on "bi bi"/"ib ib" utterances, the high-frequency energy ratio above 5 kHz is 0.15 % ("bi bi") and 0.12 % ("ib ib") at 1.10 (0.17 %/0.13 % at 0.75), with no change in F0 or duration.

## 7.3 Perceptual salience of the schwa @

With $T = 16$ and $\delta = 0.50$, the boundary schwa @ in `big.bi` is **salient** (clearly perceivable) because:

1. the C.C boundary anchor sits at $\rho = 0.45$ (most centralized, far from the full vowel /i/ at $\rho = 0.9$);
2. the $z_v$ background dips to that anchor between the /g/ and /b/ clusters (audible schwa-like vocoid);
3. there is no pause: the schwa is carried by the planning trajectory through the consonant–consonant chain, exactly as the dot semantics prescribe.

As $\delta$ increases toward 1.0, the boundary anchor approaches $V = /i/$ (COEFCEN = 1) and the perception shifts from `big.bi` to `bi.gbi`.

# 8 Code Structure

## 8.1 Modified files (vs original timit-to-Maeda)

| File | Modification |
|---|---|
| `synthSYL/constants.py` | Added `DEFAULT_DELTA_O`, `DEFAULT_DELTA_E` |
| `synthSYL/gesture.py` | Added $\delta_o$, $\delta_e$ kwargs to `build_gesture_anchors()`, `_insert_word_end_anchors()`, `_insert_vowel_onset_anchors()`, `_insert_syllable_onset_anchors()` Fixed $V_o = (\delta_o \cdot \rho_V, \theta_V)$ (was $[\delta_o, \theta_V]$) |
| `synthSYL/trajectory.py` | Word-end decay chains directly on the $V_e$ anchor (no held @; see §4.3) |
| `synthSYL/pipeline.py` | Added $\delta_o$, $\delta_e$ kwargs to `panphon_pipeline()` |

## 8.2 Simulation scripts

| Script | Purpose |
|---|---|
| `run_bigbi_polar_sweep.py` | Main sweep: big.bi → bi.gbi with polar coords |
| `run_ibbi_ushape_sweep.py` | U-shaped $T_p$ variation |
| `make_bigbi_dual_video.py` | Dual-panel video generation |
| `make_ibbi_polar_video.py` | Polar-only video generation |
| `make_article_figures.py` | Article-format Figures 1/3/4 + article-conditions planning panels |
| `embed_video_in_html.py` | Self-contained HTML (base64 video) |
| `compare_article_polar.py` | Compare with article Figure 4 |

# 9 Quick Demo

The minimal code to run the big.bi → bi.gbi simulation:

```
from synthSYL import panphon_pipeline
import vlam
import numpy as np
from scipy.io.wavfile import write

# Run the pipeline for "big bi" at delta=0.5 (salient schwa)
result = panphon_pipeline(
    "big bi",
    T=16,
    short_pause_duration_factor=1.0,
    delta_o=0.5,
    delta_e=0.5,
)

# Synthesize audio
state = vlam.VlamState.initial(195)
config = vlam.SynthConfig(play_audio=False)
sr = vlam.synthwordfen(
    state=state,
    articulatory_params=result.Pval,
```

```
    word_tokens=["O", "V", "F"],
    duration_factor=16,
    envelope=result.envelope,
    config=config,
)

# Save WAV
sig = sr.signal
sig_int16 = np.int16(32767 * sig / (1.01 * np.max(np.abs(sig))))
write("big_bi_delta05.wav", 20000, sig_int16)
print(f"Signal: {sig.shape[0]} samples ({sig.shape[0]/20000*1000:.0f} ms)")
```

# 10 Article Demonstrations

This section documents four additional demonstrations from the article that have been implemented in this repository: Demo 1 (/ibia/), Demo 2 (locus equations), Demo 3 (trough effect) and Demo 4 (consonant clusters).

## 10.1 Demo 1: /ibia/ VCV synthesis (Figure 1)

The article's Figure 1 shows the four-step synthesis pipeline on the sequence /ibia/ (/i/ → /b/ → /i/ → /a/). The four panels are:

1. **(a) Planning trajectories** $z_v(t)$ and $z_c(t)$ in the polar plane $(\rho, \theta)$, article format. The vocalic branch $z_v$ (red) flows from /i/ to /i/ to /a/, while the consonantal branch $z_c$ (blue) makes a teardrop excursion toward the /b/ target.
2. **(b) Flow of articulatory parameters** — the 7 Maeda parameters over time. The *Body* parameter makes an unexpected front/back movement during /b/, rising toward the /u/ direction (numerically a rise; articulatorily a lowering and backing of the tongue body away from /i/, per the Maeda sign convention — see Demo 3), because /b/ lies on the same angular direction as /u/ in the complex plane. With the synthSYL registry default $\rho_b = 1.2$ used by this figure, the Body peaks at +1.50 (measured) above the /i/ plateau −2.25; the article position $\rho_b = 1.0$ gives +1.25.
3. **(c) Formant trajectory** F1-F2-F3. F2 rises from its initial (neutral-onset) value (≈ 1830 Hz, measured 1826 Hz at frame 0) to the /i/ plateau (≈ 2280 Hz, measured 2276 Hz), dips during /b/ in a V-shaped valley (minimum 1570 Hz at $\rho_b = 1.2$, mid-closure), rises back to the /i/ level, then falls toward /a/. This is a V-shaped perturbation, not an S-shape: the F2 pattern through the consonant is a single minimum between two vowel plateaus.
4. **(d) Output spectrogram + wave** showing the acoustic result (waveform superposed on the grayscale spectrogram).

## 10.2 Demo 2: Locus equations (Figure 3)

The article's Figure 3 shows the locus equations for /b/, /d/, /g/ across 8 vowels. A locus equation is the linear relationship:

$$F2_{\text{locus}} = a \cdot F2_{\text{vowel}} + b \tag{5}$$

where $F2_{\text{locus}}$ is measured 30 ms after the consonant **release** — the article: "taking the F2 values 30 ms after the onset" ($\delta_o = 0.5$), "onset" meaning the onset of the vowel at the release, in the locus-equation tradition of F2-onset measurement. In the synthesis timeline of a CV word ($T = 16$ steps of 10 ms), the release is the first frame of the C→V transition block (frame $2T$): frames $0..T-1$ are the initial arc (neutral → Vo), frames $T..2T-1$ the Vo→C closure (envelope silent), frames $2T..3T-1$ the C→V transition (envelope rising), and frames $3T..4T-1$ the vowel plateau. Hence $F2_{\text{locus}} = F2$(frame $2T+3$) and $F2_{\text{vowel}}$ is averaged over the plateau frames (envelope at full amplitude). The script (`scripts/run_locus_trough_demos.py`) detects the release and the plateau from the envelope; the offset (`LOCUS_OFFSET_STEPS = 3` in `config.py`) is the article's 30 ms.

/g/ uses two article positions: velar ($\rho = 1.2, \theta = \pi/3$) for the back vowels {/u/, /o/, /O/} and palatal ($\rho = 1.1, \theta = 23\pi/12$) for the front vowels — *including /a/*, whose red point lies on the palatal branch in the article's Figure 3.

**Results (corrected protocol):**

| Consonant | Slope ($a$) | Intercept ($b$) | $r^2$ | Article Fig. 3 ($a$) |
|---|---|---|---|---|
| /b/ | 0.703 | 357 Hz | 0.983 | $\approx 0.70$ |
| /d/ | 0.421 | 1245 Hz | 0.965 | $\approx 0.55$ |
| /g/ palatal (front + /a/) | 0.630 | 916 Hz | 0.784 | $\approx 0.75$ |
| /g/ velar (back) | 2.335 | −1501 Hz | 1.000 | $\approx 2.0$ |

Measured at `valrect` = 1.10 (a documented difference; the article publishes no valrect — 0.75 is the historical timit-to-Maeda default; see §7.2 and the Differential Supplement, §4, row 6). The velar $r^2 = 1.000$ is computed on only 3 points (the back vowels o, O, u) and is therefore barely informative.

The article-figure slopes in the last column were measured by pixel extraction from the arXiv source figure (marker centroids calibrated on the axis ticks). The /b/ slope matches the article value (≈0.70), and the /g/ split (steep velar, shallow palatal) is qualitatively reproduced. The remaining slopes deviate quantitatively from the article: /d/ is flatter here (0.42 vs ≈0.55), the palatal /g/ shallower (0.63 vs ≈0.75) and the velar /g/ steeper (2.34 vs ≈2.0). What is preserved is the ordering and the classical pattern: $a(/d/) < a(/b/) < 1$, the /d/ locus at back vowels (1690–1782 Hz) in the canonical alveolar range (≈1700–1800 Hz), and the slope ordering of the locus-equation literature, where slopes descend from labial to velar to alveolar and the alveolar line is the flattest with the highest intercept [2, 3]. The article notes: *"This is similar to the classical observation but with a downward shift of the velar /g/ which could be due to geometric differences between the VLAM and the human vocal tract."* The velar points here indeed fall on or below the $y = x$ diagonal (/g/+u: 1029 → 966 Hz).

**Why the slopes deviate from the article.** A locus slope is a RATIO, $a = \Delta F2_{\text{locus}}/\Delta F2_{\text{vowel}}$ across the 8 vowels, so it depends on the SPREAD of the $F2_{\text{vowel}}$ axis, not only on the consonant. The $F2_{\text{vowel}}$ axis here is compressed relative to the article's figure (e.g. /i/: 2290 Hz here vs ≈2660 Hz, /u/: 1029 vs ≈880): the article was produced with an earlier synthSYL vowel table (corner vowels at $\rho = 1$ per §2.1), which has since been revised in `timit-to-Maeda`. A slope is fitted as a ratio against this axis, so the compression changes every fitted value, and the line with the highest intercept is the most sensitive — the /d/ line (intercept ≈ 1250 Hz, the flattest of the three) moves the most (0.42 here vs ≈0.55 in the article), while the ordering and the linearity of the points are essentially unaffected. This is a coordinate-table difference, not a change in the coarticulation behaviour.

### 10.2.1 Measurement-protocol sensitivity

The measured slopes depend on the protocol, and the sensitivity was checked on three factors:

- **The soft rectification is not implicated.** `valrect` (`soft_rect_s`) is applied to the *area function* (soft_rect of section areas, preventing zero/negative areas during closure), not to the articulatory parameters. A sensitivity sweep shows the locus slopes are insensitive to it over $s \in [0.5, 2.0]$ ($a(/d/) = 0.562 \pm 0.011$ at release+50 ms); only the hard limit $s \to 0$ (hard clipping) breaks the synthesis.
- $\delta_o$ **is not implicated either.** Sweeping $\delta_o \in \{0.5, 0.7, 1.0\}$ moves $a(/d/)$ by less than 0.02.
- **The measurement instant is the decisive parameter.** With the release located at frame $2T$, the /d/ slope varies from 0.25 (release+0) to 0.79 (release+6 frames), passing through the article value around release+3 (30 ms). The /b/ slope varies from 0.50 to 0.92 over the same range.

The measurement logic lives in the demo script (`scripts/run_locus_trough_demos.py`) and `config.py` (constant `LOCUS_OFFSET_STEPS` = 3, the article's 30 ms); no engine file (`vlam.py`, `synthSYL/`) is involved.

**Note:** For a detailed comparison with Ohman's spectrographic data, see the 2025 companion article (arXiv:2505.22455).

## 10.3 Demo 3: Trough effect (Figure 1b)

The article (§3) describes an unexpected articulatory phenomenon during /b/:

> *"The tongue body is making an unexpected front/back movement during /b/ which corresponds to the trough effect [29]."*
>
> *(reference [29]: Lindblom, Sussman, Modarresi Ghavami & Burlingame,* Phonetica *59:245–262, 2002)*

**The source.** Lindblom et al. [29] trace the effect to Perkell [1986, p. 48], who described it as an "apparent discontinuity in anticipatory coarticulation": in VCV utterances with identical vowels around a labial stop (e.g. [ibi]), the muscle activity maintaining the tongue configuration for V1 is diminished ("turned off") into the stop closure, the tongue dorsum lowers, and returns to the same vowel target as V2 approaches (their Introduction). Two properties of the source matter for the comparison below: the reported movement deviations are "relatively small" and highly variable, and the effect has been documented *only* from movement-related data (X-ray, EMG, electropalatography) — never acoustically.

**The model.** In the present implementation, /b/ lies on the **same angular direction as /u/** in the complex plane. The article value is confirmed in the arXiv source (§3, verbatim): *"we found that /b/ is easy to reach with* ($\rho_b = 1,\ \theta_b = \pi/3$)*"*. The code constants agree: the vowel table of `synthSYL/phonology.py` places /u/ at ($\rho_u = 1.0,\ \theta_u = \pi/3$), so at the article position the two targets coincide exactly ($\rho_b = \rho_u$). The synthSYL registry default $\rho_b = 1.2$ shares the same direction but a larger radius; the article-format Figure 1 keeps this default, which is why its Body peak is +1.50 instead of +1.25 (see below). During the /b/ consonant the Body parameter rises from the vowel plateau toward the /u/-direction value — a **numerical peak**, not a dip, whatever the vowel context (verified by script; see the table below). This increase

follows the Maeda SIGN CONVENTION: negative Body values encode the fronted/high tongue-body posture of /i/, positive values a more retracted posture. The rise from $-2.25$ to $+1.25$ therefore corresponds articulatorily to a POSTERIOR and DOWNWARD displacement of the tongue body relative to the extreme front/high posture of /i/ — a lowering (and backing) of the tongue with respect to V1, not an elevation.

**How Body is actually computed (resolves the $S_v/S_c$ partition).** The selection vectors are *not* a partition of the 7 parameters into vowel-driven and consonant-driven subsets. Body belongs to BOTH $S_v$-related (all 7 parameters are written by the vowel anchor at plateaus) and $S_c = \{1, 2, 6\}$ (code $\{0, 1, 5\}$). What happens is a *sequential overwrite* (see `synthSYL/trajectory.py`, `inject_active_parameters()`, lines 156–167): vowel plateaus write all 7 parameters from $z_v$ (giving the /i/ plateau value Body $= -2.25$), then the cluster block overwrites the $S_c$ parameters from the consonantal arc $z_c$ (giving the /u/- direction value Body $= +1.25$ at $\rho_b = 1.0$). The "vowel articulators stay around /i/" of the article holds because the NON-overwritten parameters (Dorsum, Tip, LipP, Hy, i.e. $S_v = \{3, 4, 5, 7\}$ article numbering / code $\{2, 3, 4, 6\}$) keep their $z_v$ values through the cluster.

**Measured values** (from the recorded engine blocks; article positions $\rho_b = 1.0$; T = 16; the full table is reproduced by `scripts/run_trough_ablations.py`):

| VCV | Plateau Body | Extremum during C | Instant | Amplitude |
|---|---|---|---|---|
| /ibi/ | $-2.250$ | $+1.250$ (max) | 630 ms | $+3.500$ |
| /aba/ | $+1.000$ | $+1.298$ (max) | 590 ms | $+0.298$ |
| /idi/ | $-2.250$ | $-2.228$ (max) | 580 ms | $+0.022$ |
| /igi/ | $-2.250$ | $-1.945$ (max) | 630 ms | $+0.305$ |
| /ibia/ (T=10) | $-2.250$ | $+1.250$ (max) | 390 ms | $+3.500$ |

All Body values are Pval-level (planning/engine) quantities and are therefore valrect-independent; the formant values in this section are measured at `valrect` = 1.10 (§7.2). The medial closure spans frames 61–64 and the release is at frame 65 (650 ms) for the T=16 words; the extremum always falls INSIDE the cluster block (e.g. 630 ms for /ibi/, whose cluster spans frames 48–79 = 480–790 ms). In /ibi/, F2 falls from the /i/ plateau 2276 Hz to a V-minimum of 1630 Hz at 630 ms and returns to 2276 Hz.

The excursion is a **numerical rise toward** $+1.25$ peaking at $\approx 630$ ms (T=16) or $\approx 390$ ms (T=10) — it must be measured inside the recorded cluster blocks against the preceding vowel plateau; a naive scan of frames $T..3T$ would instead pick up the /i/ plateau itself (Body $= -2.25$). In articulatory terms, and recalling the sign convention above, this numerical peak is a lowering and backing of the tongue body relative to /i/.

**Why $+1.25$ here and $+1.50$ in Figures 1/10–12.** The amplitude depends on $\rho_b$: the registry default $\rho_b = 1.2$ (used by the article-format Figure 1 and the planning panels) gives Body $= +1.500$, while the article position $\rho_b = 1.0$ (set explicitly by `run_locus_trough_demos.py`) gives $+1.250$ (both measured). The plateau value $-2.25$ is identical in both configurations.

**Ablations** (`scripts/run_trough_ablations.py`, /ibi/, T=16):

| Configuration | Body excursion | F2 minimum | F2 dip |
|---|---|---|---|
| baseline ($S_c = \{0, 1, 5\}$, $\rho_b = 1$, $\theta_b = \pi/3$) | +3.500 | 1630 Hz | yes |
| (i) Body removed from $S_c$ ($\{0, 5\}$) | +0.003 | 2273 Hz | no |
| (ii) $\rho_b = 0.5$ | +2.875 | 1773 Hz | yes |
| (ii) $\rho_b = 1.0$ | +3.500 | 1630 Hz | yes |
| (ii) $\rho_b = 1.2$ | +3.750 | 1570 Hz | yes |
| (iii) $\theta_b = \pi$ | +3.500 | 1582 Hz | yes |
| (iii) $\theta_b = 3\pi/2$ | +0.086 | 2241 Hz | no |
| (iv) /d/ (idi) | +0.022 | 2130 Hz | small |
| (iv) /g/ palatal (igi) | +0.305 | 2269 Hz | no |

What these measurements show and nothing more: the excursion exists because Body belongs to the /b/ selection vector (i); its amplitude scales with $\rho_b$ (ii); with $\theta_b$ far from the /u/ AND /i/ directions ($3\pi/2$) the excursion vanishes, while at $\theta_b = \pi$ the Body amplitude is unchanged (the Body value responds mostly to $\rho$, the FORMANTS to $\theta$ — the F2 minimum differs between $\theta_b = \pi$ and $\pi/3$); and among /b d g/ only /b/ (and weakly /g/) produces it (iv).

**Emergence without deactivation.** The model implements no explicit activation/deactivation mechanism: the trough-like pattern emerges from the continuous modulation of the two trajectories $z_v(t)$, $z_c(t)$ combined with the selection vectors. It can therefore be regarded as a *functional analogue* of the deactivation described by Lindblom et al. — obtained without taking activation/deactivation as a primitive of the control architecture. This is an interpretive consequence of the present model, not a claim of the original article.

**Relation to the source.** Read with the sign convention in mind, the kinematic pattern produced by the model is more consistent with Lindblom's trough effect than the purely numerical "peak" wording suggests: what the simulation produces during the labial stop is a lowering and backing of the tongue body away from the V1 posture (toward the /u/ direction) — precisely the kinematic signature of the trough, the dorsum leaving the V1 configuration into the closure and returning as V2 approaches. It is not, however, a confirmation of the Lindblom mechanism: (a) the direction is a consequence of the $\theta_b = \theta_u$ model placement, which is itself a modeling choice; (b) the simulated amplitude (+3.50 parameter units over /i/) is an order of magnitude larger than the "relatively small" deviations reported in the articulographic literature — an architectural consequence of Body $\in S_c$ — to be discussed if a quantitative comparison is ever established; (c) the original observations rest on movement data only, never acoustics; (d) the sign of the simulated excursion follows the context (a weak bump +0.30 over /a/, whose Body level +1.00 already sits near the /u/ direction).

An interactive HTML simulation illustrating this effect accompanies this section (directory `simulations/`).

## 10.4 Demo 4: Six consonant cluster pairs

The article (§3) describes the tuning of the six pairs of /b,d,g/:

| **Pair** | **Input** | **Sc** | **Note** |
|---|---|---|---|
| /bd/ | abda | {1,2,3,6} | /b/ involved → Sc={1,2,3,6} |
| /bg/ | abga | {1,2,3,6} | /b/ involved, /g/ palatal |
| /db/ | adba | {1,2,3,6} | /b/ involved → Sc={1,2,3,6} |
| /dg/ | adga | {1,2,3} | engine threshold rule (place-index sum > 2) |
| /gb/ | agba | {1,2,3,6} | /b/ involved, /g/ palatal |
| /gd/ | agda | {1,2,3} | engine threshold rule (place-index sum > 2) |

**Rule (engine, verified):** the selection vector is $S_c = \{1, 2, 3, 6\}$ (code $\{0, 1, 2, 5\}$ = jaw, body, dorsum, LipH) when the sum of place indices ($b = 0$, $d = 1$, $g = 2$) is $\leq 2$, and $S_c = \{1, 2, 3\}$ (code $\{0, 1, 2\}$ = jaw, body, dorsum) otherwise. Hence /bb/, /bd/, /bg/, /db/, /dd/, /gb/ use $\{1, 2, 3, 6\}$ and /dg/, /gd/, /gg/ use $\{1, 2, 3\}$. The article's rule — $\{1, 2, 3, 6\}$ for clusters containing /b/ (e.g. /gb/), $\{1, 2, 3, 4\}$ otherwise — differs from the engine's for the four pairs /dd/, /dg/, /gd/, /gg/. The engine's rule is inherited from the 2023 lineage: both original engines implement the same threshold rule; the article's /b/-conditional rule was never implemented in the family. The /g/ is systematically palatal when in a cluster (see the Differential Supplement, §4, rows 4 and 4a).

**Labels note.** The $S_c$ annotations in the figures above are derived at runtime from the engine (`synthSYL.phonology.cluster_art_params`, article numbering) and match the rule paragraph above; the trajectories are unaffected. The spectrogram titles carry no $S_c$ label. See the Differential Supplement, §4 row 4 and Corrigendum 3, for the history of the earlier legacy labels.

### 10.5 Figure 4: big.bi | bi.gbi (article format)

big.bi is rendered with the **dot form** (`big.bi`): per the reference `Timit-to-Maeda` semantics (`_should_split_at_dot`: C.C → syllable split, C.V / V.C / V.V → merge), the dot between /g/ and /b/ is a syllable boundary with *no pause*, and the Ve of "big" is coarticulated with the /i/ of "bi" (syllable-onset anchor weighted by COEFCEN; full /i/ under COEFCEN = 1). Under the article figure conditions the left column thus shows three consonantal gestures (/b/; then /g/ and /b/ chained across the shared /i/) while the right column (bi.gbi, whose V.C dot is ignored) shows the fused /gb/ cluster — the §4 syllabification demonstration.

Note the two parameter conventions: this figure fixes the ARTICLE figure conditions $\delta_o = 0.5$ (VOYDEB), $\delta_e = 1$ (COEFCEN), whereas the Demonstration 1 sweep varies a single $\delta = \delta_o = \delta_e$ (COEFCEN) from 0.5 to 1.

The supplement transformation /ib/ → /bi/ (§3: "The classical transformation of /ib/ into /bi/ is given in the supplement.") is reproduced by `scripts/make_supplement_ib_bi.py` (→ `docs/figures/article_format/supplement_ib_bi.png`): the three cases /ib/ $\delta = 0.5$ (schwa @ audible, /b/ in coda), /ib/ $\delta = 1.0$ ($V_o = V_e = V$, fusion) and /bi/ $\delta = 1.0$ (/b/ in onset, the fused form), each as Maeda parameters + formants + spectrogram. Gate: /ib/ is 112 steps = 1120 ms at BOTH deltas, the /b/ cluster followed DIRECTLY by the decay (no held @). The decay is a T-step hold at $V_e$ with the envelope closing (plus the silent terminal arc), so the visible stay at $V_e$ ($\approx 2T$) is exaggerated — deliberately, so that the anchoring is demonstrable.

## 11 Interactive HTML Simulations

The repository includes three self-contained HTML files (in the `simulations/` directory) with video, poster, captions, and comparison figures embedded as base64 data URIs. These files can be opened directly in any modern browser without requiring a web server or external file dependencies.

### 11.1 big.bi → bi.gbi (T=16, starts at COEFCEN=0.5)

`simulations/bigbi_bi_gbi_embedded.html` (≈ 6.3 MB)

This simulation demonstrates the verbal transformation from big.bi to bi.gbi (§4 of the article), in the **dot form** (no pause, no $T_p$ dimension). It starts at $\delta = 0.50$ (COEFCEN = 0.5, salient

boundary schwa @) then increases $\delta$ to 1.00 (the C.C boundary anchor reaches the full /i/, fusion). The last bi.gbi segment is repeated 3 times for perceptibility. Duration ≈ 24.2 s.

Features: dual-panel video (sagittal Maeda | polar plot with phoneme), 14 segments, WebVTT captions, indexed transcript, keyboard shortcuts.

### 11.2 ib ib → bi bi (NON-REVERSIBLE U-shape)

`simulations/ibbi_ushape_nonreversible_embedded.html` (≈ 4.4 MB)

This simulation demonstrates the **non-reversible** verbal transformation with a **U-shaped $T_p$ like the reversible demonstration** (but not with the identical values: the descent passes through $T_p = 10$ ms before the fusion point, whereas the reversible sweep jumps 40 → 0 ms): "ib ib" (VC.VC, /b/ in coda) with $T_p$ decreasing 160, 120, 80, 40, 10 ms ($\delta$ correlated at every point); at the fusion point ($T_p = 0$, $\delta = 1$) the input switches to "bi bi" (CV.CV, /b/ in onset); then $T_p$ increases again 40, 80, 120, 160 ms — but the content STAYS "bi bi": the /b/ remains in onset position even with a fully restored pause, proving the transformation is permanent and non-reversible. Duration ≈ 16.9 s (10 segments; the reversible sweep is 18.6 s).

Features: dual-panel video, 10 segments, WebVTT captions, indexed transcript, keyboard shortcuts.

### 11.3 ib ib → bi bi (U-shape, reversible)

`simulations/ibbi_ushape_embedded.html` (≈ 5.3 MB)

The reversible version: a **continuous U-shaped ramp** of 10 segments (see Demonstration 2) — "ib ib" throughout, with $T_p$ 160 → 0 → 160 ms, $\delta$ correlated. This version returns to "ib ib" at the end (reversible). Included for comparison with the non-reversible version above. Duration ≈ 18.6 s.

Features: dual-panel video, 10 segments, WebVTT captions, indexed transcript, keyboard shortcuts.

## 12 Conclusion

This repository provides a complete, reproducible implementation of the verbal transformation described in arXiv:2307.02299. The key contributions are:

1. The $\delta_o$ and $\delta_e$ parameters, enabling the gradual transition from schwa-like vocoid to full vowel.
2. The syllable-onset anchor weighted by COEFCEN (the reference Timit-to-Maeda formula), and no held @: the trajectory reaches the word-end anchor and decays immediately (one does not pronounce "ibe").
3. The $V_o$ formula correction ($V_o = \delta_o \cdot \rho_V$), ensuring $V_o = V$ at $\delta = 1$ for proper fusion.
4. The polar trajectory reconstruction, reproducing the two-branch visualization (vocalic $z_v$, consonantal $z_c$) described in the covtl-pipeline documentation.
5. The U-shaped $T_p$ variation, for clear perceptual demonstration of the ib ib → bi bi transformation.

The duration reduction observed in the dot form is $1T = 160$ ms (with $T = 16$), from consonant clustering alone (§7.1), matching the article's "the gain of consonant clustering is a reduction of one period $T$". (The article's broader $3T$ decomposition additionally involves the inter-word pause machinery of the space form, which the dot form does not have.)

The polar topology matches the article's Figure 4: at $\delta < 1$, the consonantal branch makes 2 separate excursions; at $\delta = 1$ (bi.gbi), it makes 1 fused /gb/ excursion.

## References

[1] F. Berthommier, *Why can big.bi be changed to bi.gbi? A mathematical model of syllabification and articulatory synthesis*, arXiv:2307.02299, July 2023.

[2] H. M. Sussman, H. A. McCaffrey and S. A. Matthews, *An investigation of locus equations as a source of relational invariance for stop place categorization*, J. Acoust. Soc. Am. 90(3):1309–1325, 1991.

[3] K. Iskarous, C. A. Fowler and D. H. Whalen, *Locus equations are an acoustic expression of articulator synergy*, J. Acoust. Soc. Am. 128(4):2021–2032, 2010.

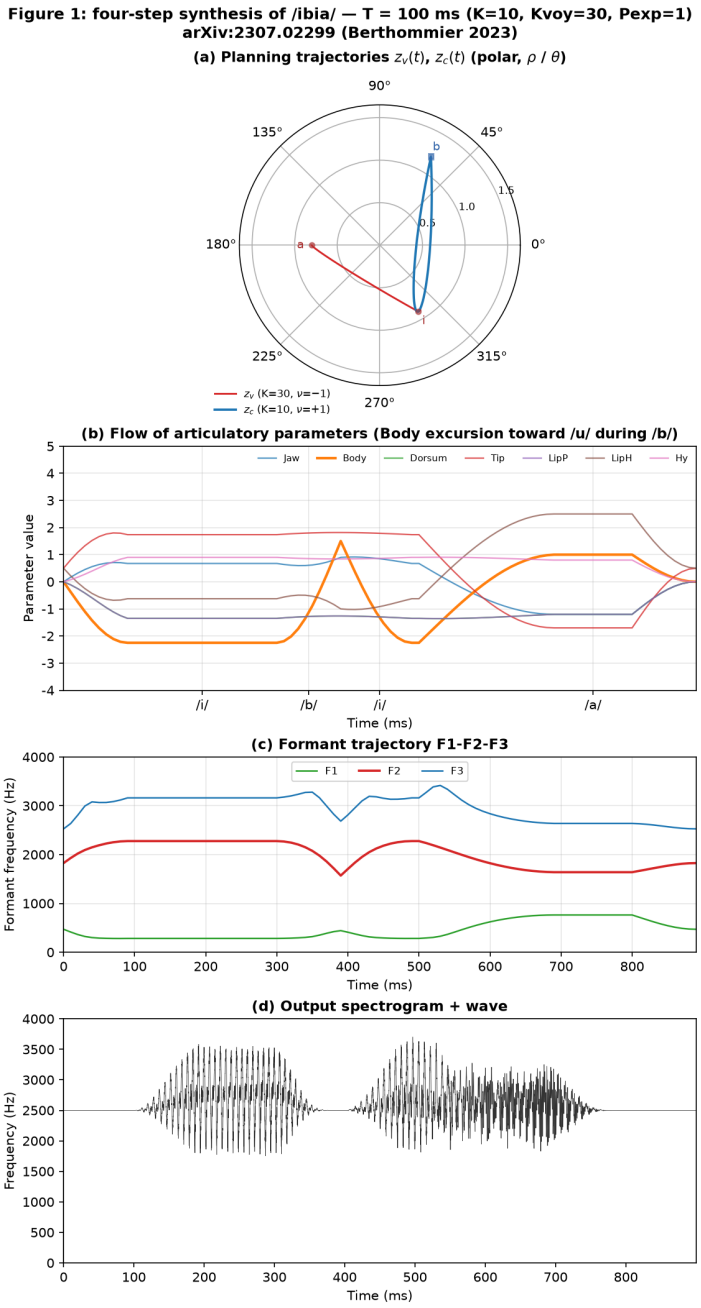



Figure 1: Figure 1, article format (`make_article_figures.py`): four-step synthesis of /ibia/ with $T = 100$ms ($T_{\text{base}} = 10$, K=10, Kvoy=30, Pexp=1). (a) Polar planning trajectories ($z_v$ red, $z_c$ blue — original arcplot anchoring, teardrop $z_c$). (b) The 7 Maeda parameters with phoneme ticks /i/ /b/ /i/ /a/ (Body excursion toward /u/ during /b/). (c) Formant trajectory (V-shaped F2 dip during /b/). (d) Spectrogram + wave.

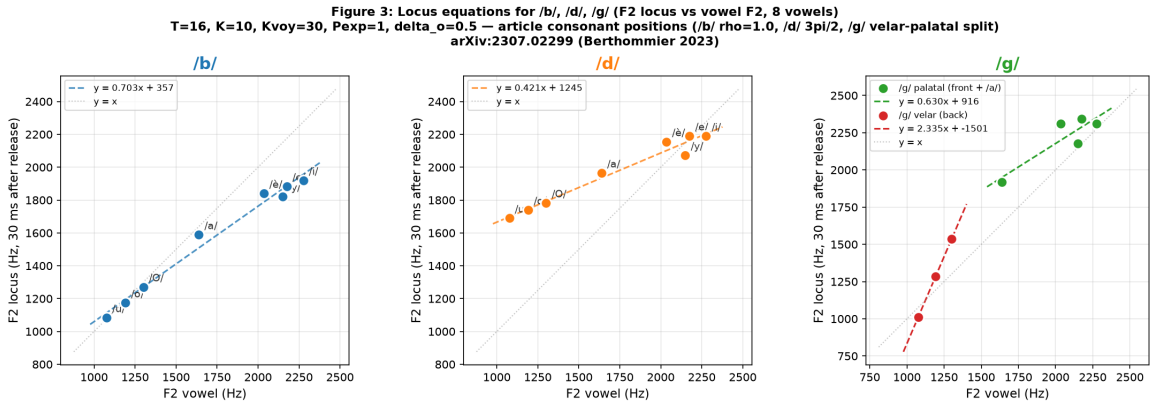



Figure 2: Figure 3, article format (`make_article_figures.py`): locus equations for /b/, /d/, /g/ across 8 vowels. Each panel shows $F2_{\text{locus}}$ vs $F2_{\text{vowel}}$ with the linear regression line.

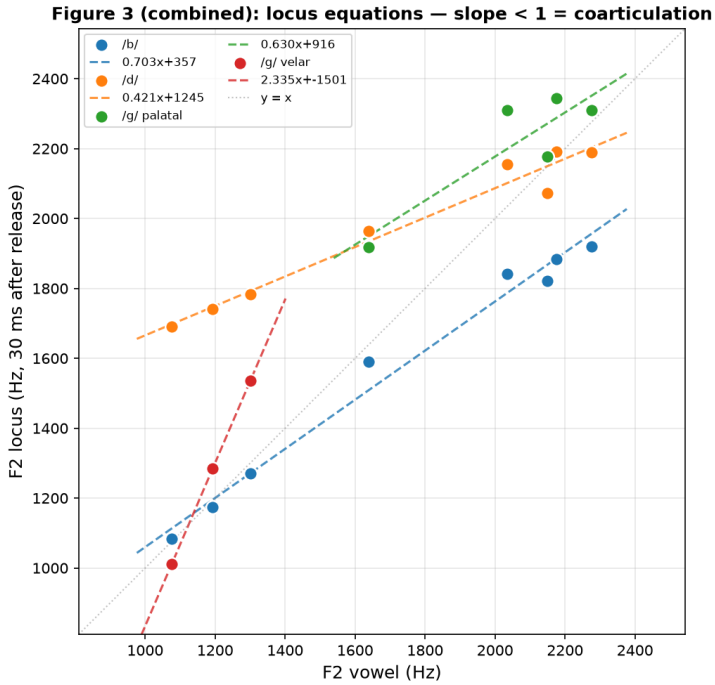



Figure 3: Figure 3 (combined), article format: locus equations for all three consonants. The dotted line ($y = x$) represents no coarticulation. Slopes $< 1$ indicate coarticulation (the consonant locus is influenced by the vowel context); the velar /g/ slope $> 1$ reflects the opposite regime (locus extension toward the back vowels), as in the article.

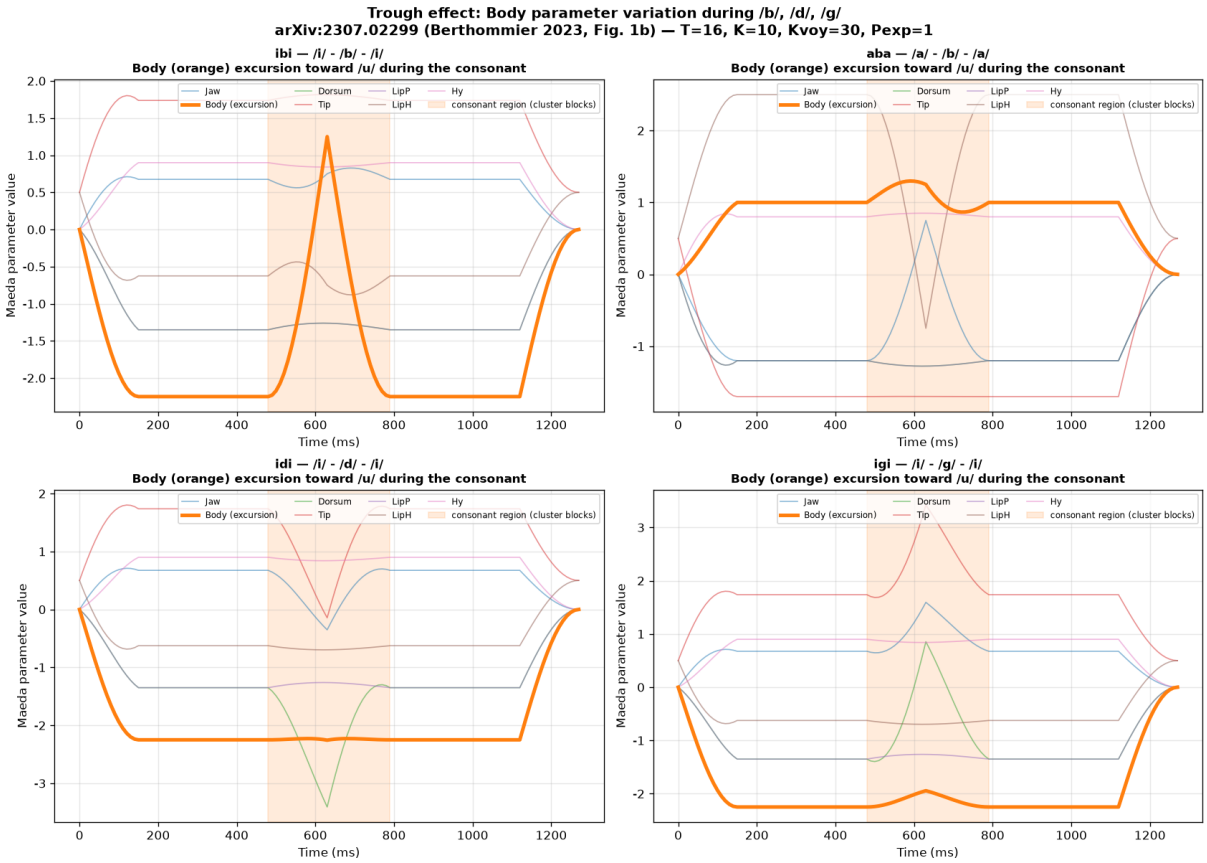



Figure 4: Trough effect: Body parameter variation during /b/, /d/, /g/ in VCV sequences. The Body (orange, thick line) shows a numerical PEAK during the consonant for /ibi/ (rise from the /i/ level −2.25 toward the /u/-direction value +1.25; articulatorily, per the Maeda sign convention, a lowering and backing of the tongue body away from /i/); the excursion over /aba/ is weak (+0.30). The shaded region is the recorded engine cluster block (frames 480–790 ms at T=16).

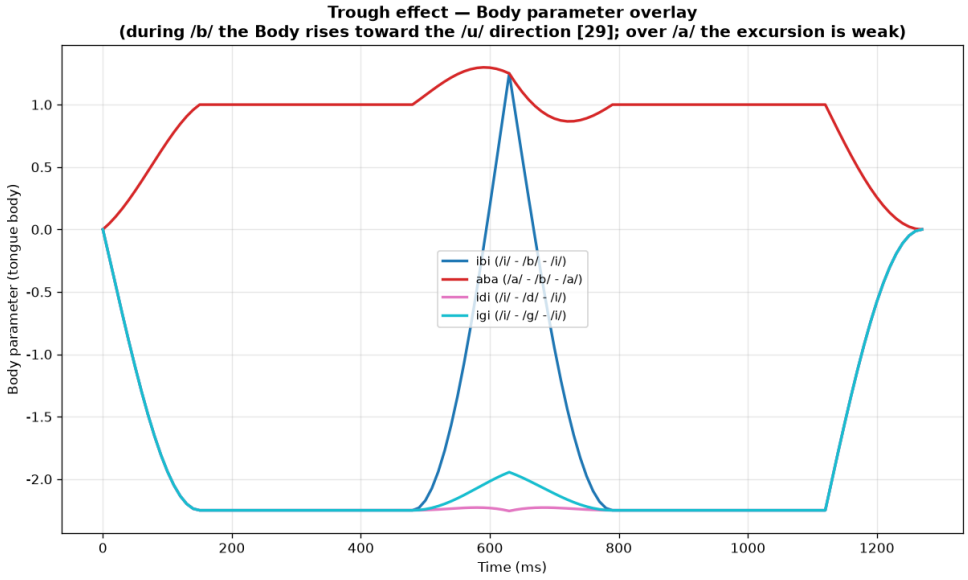



Figure 5: Body parameter overlay for /ibi/, /aba/, /idi/, /igi/. The excursion is strong for /ibi/ (+3.50 from the /i/ plateau) and weak for /aba/ (+0.30 from the /a/ plateau, whose Body level +1.00 already sits near the /u/ direction); /idi/ and /igi/ are nearly flat. The /b/ target lies on the same angular direction as /u/; the numerical rise corresponds to a lowering/backing of the tongue body relative to /i/ (Maeda sign convention).

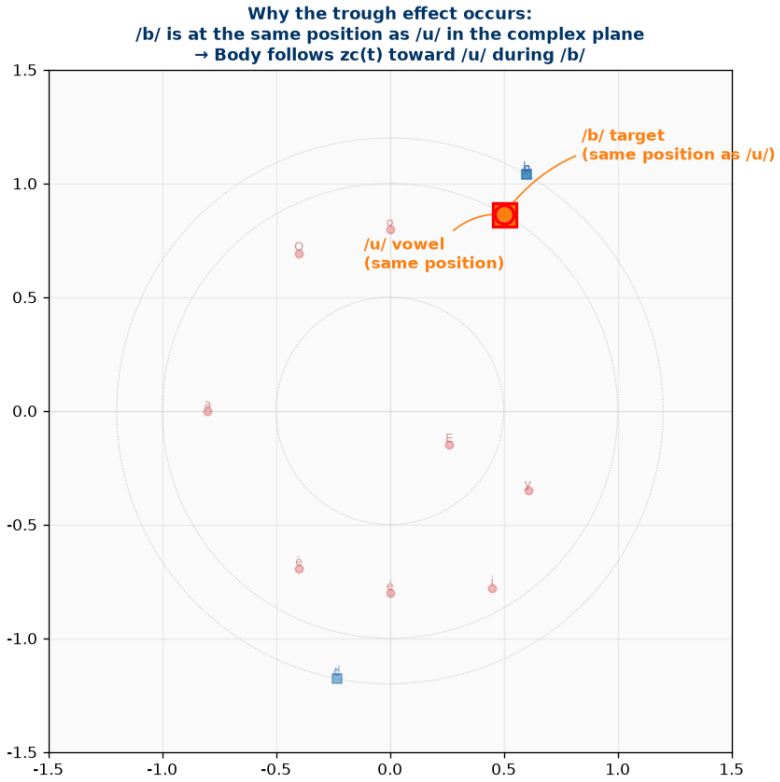



Figure 6: Polar explanation: /b/ (orange square) lies on the same angular direction as /u/ (orange circle) in the complex plane — with the article position $\rho_b = 1$ the two targets coincide. During /b/, the Body parameter follows $z_c(t)$ toward /u/, producing the excursion — a tongue-body lowering/backing relative to /i/ — while the non-overwritten vowel articulators stay around /i/.

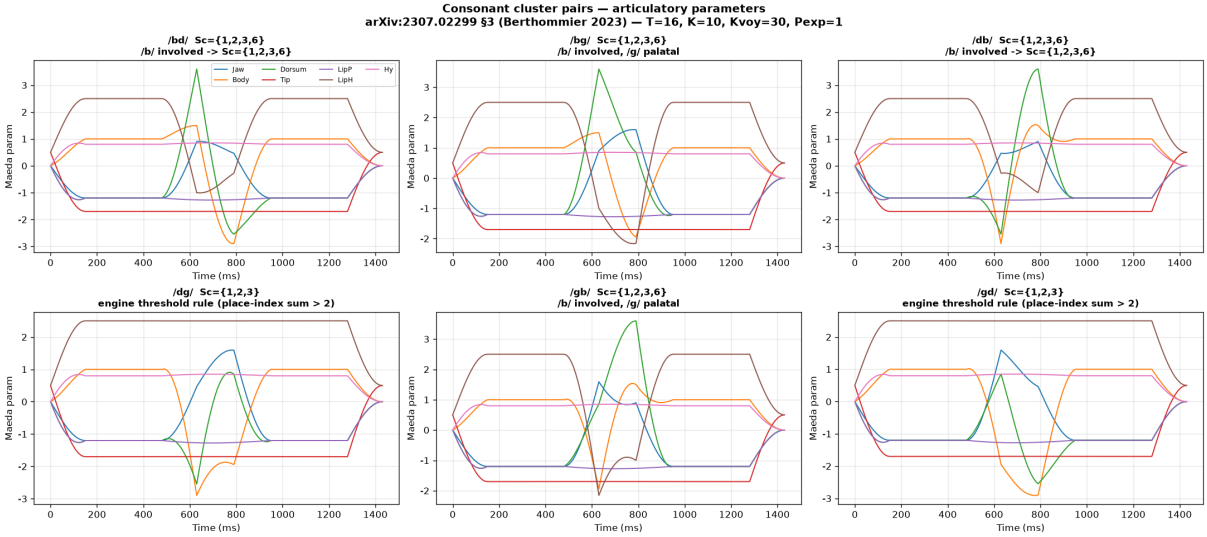



Figure 7: Articulatory parameters for the six cluster pairs. Each panel shows the 7 Maeda parameters over time.

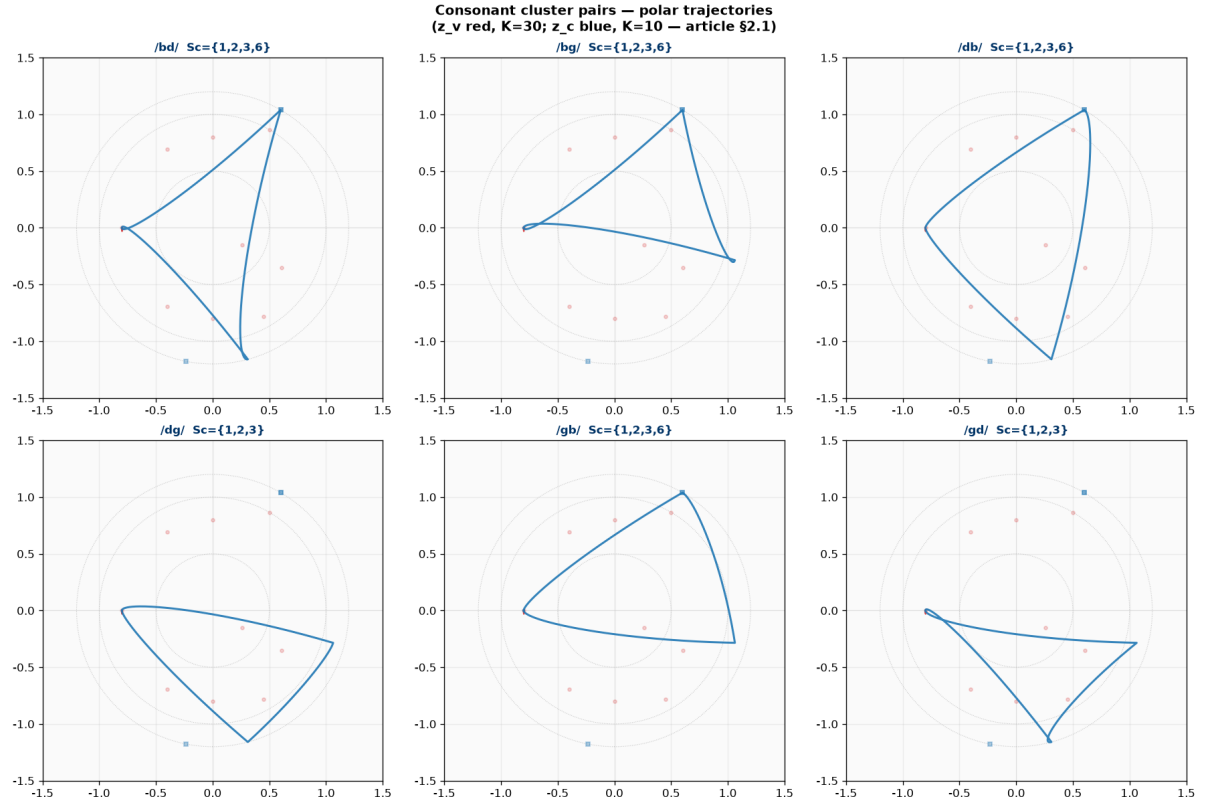



Figure 8: Polar trajectories for the six cluster pairs. The consonantal branch $z_c$ (blue) visits both consonant targets in sequence.

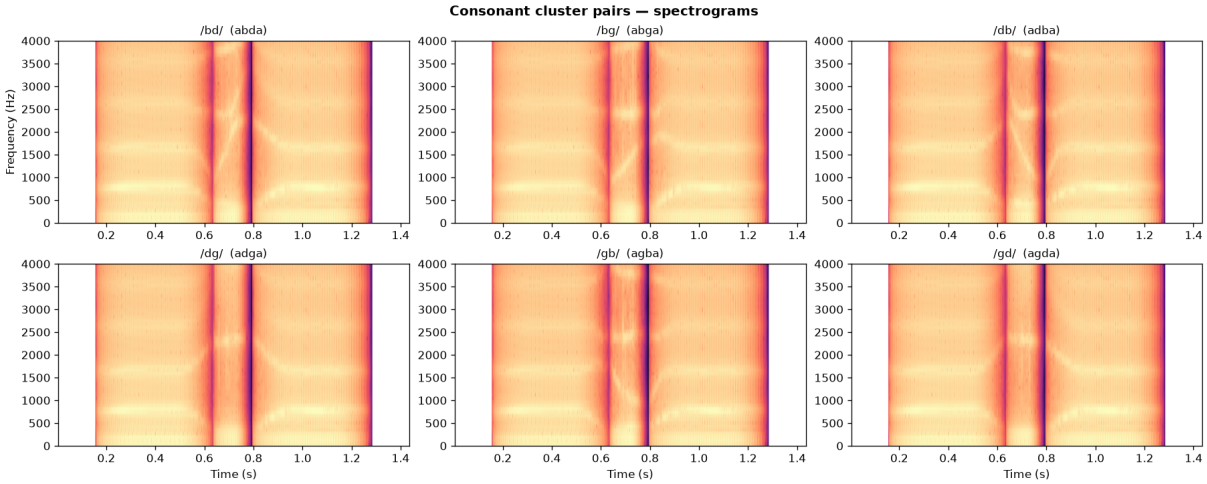



Figure 9: Spectrograms for the six cluster pairs.

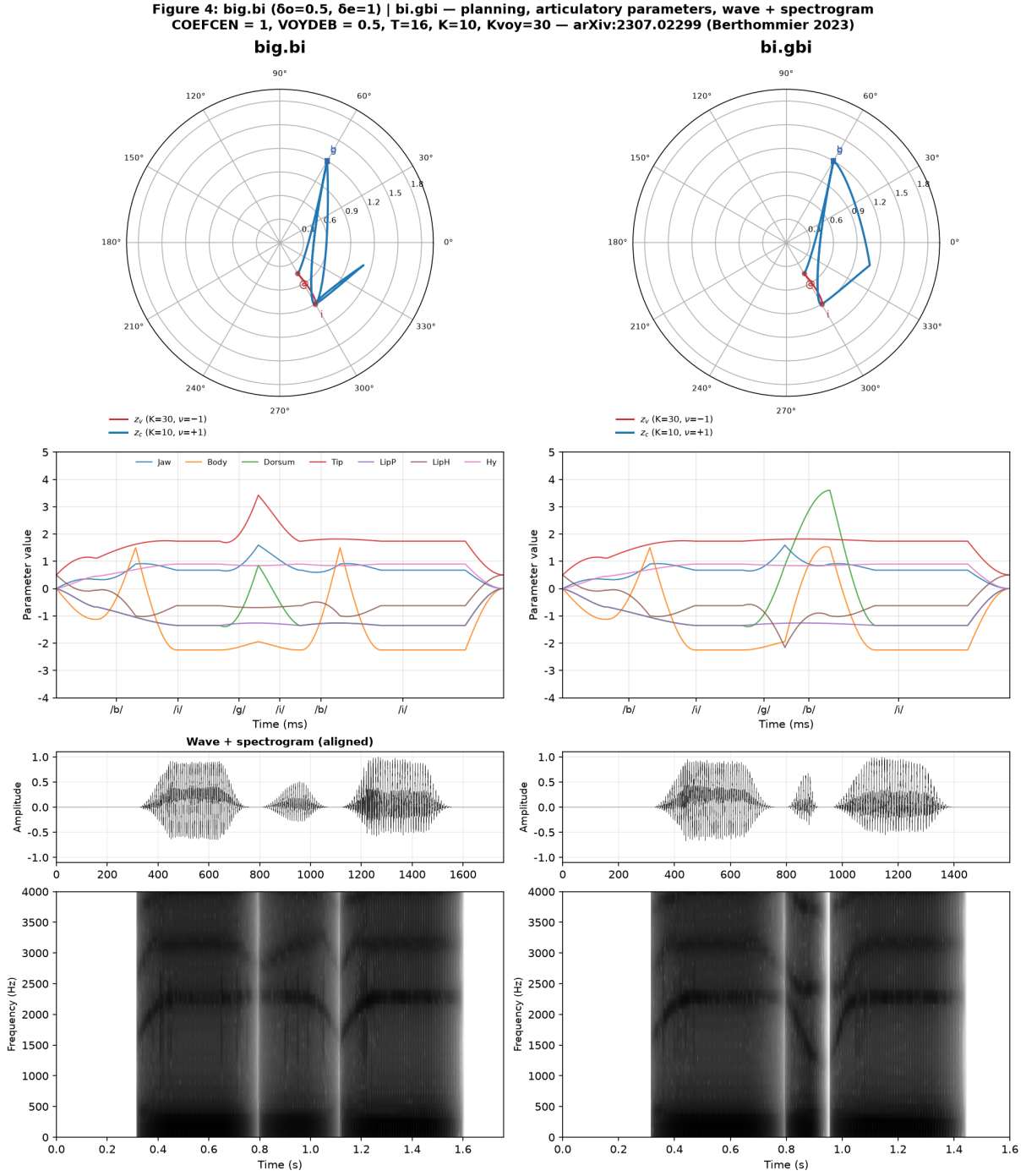



Figure 10: Figure 4, article format (`make_article_figures.py`): big.bi | bi.gbi under the article figure conditions COEFCEN = 1 ($\delta_e$), VOYDEB = 0.5 ($\delta_o$), $T = 16$, both in the **dot form** (no pause). big.bi is the C.C syllable concatenation: the Ve of "big" is coarticulated with the /i/ of "bi" (the syllable-onset anchor is the full previous vowel under COEFCEN = 1) — the only silences are the stop closures ($\approx T/2$), as in the reference realization; $z_c$ carries 3 consonantal gestures (/b/; /g/ then /b/ chained across the shared /i/). bi.gbi: the dot between /i/ and /g/ is V.C and therefore ignored (syllables merged), giving the fused /gb/ cluster (2 gestures). Top: polar planning panels (teardrop $z_c$, red $z_v$ holding the /i/ radius). Middle: Maeda parameters with phoneme ticks. Bottom: wave (ms) + spectrogram (s), same total duration.

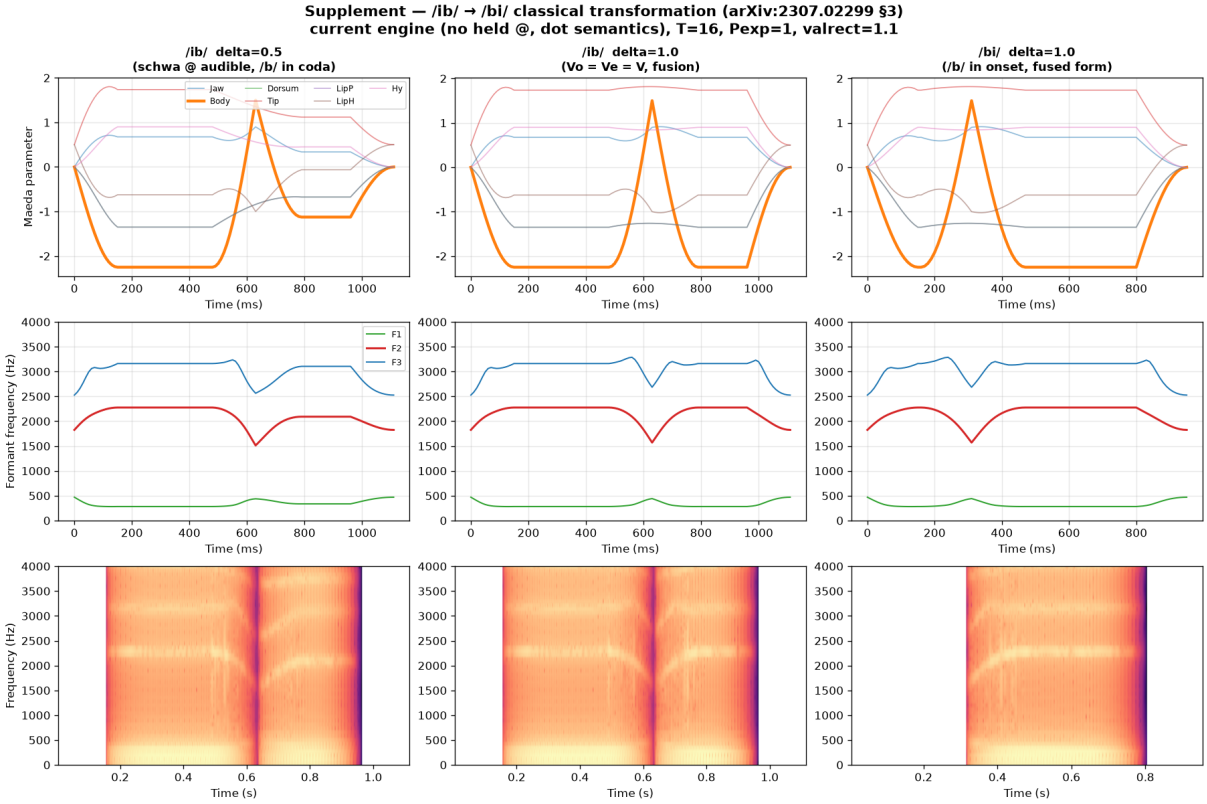



Figure 11: Supplement /ib/ → /bi/, current engine (no held @, valrect = 1.10): /ib/ $\delta = 0.5$, /ib/ $\delta = 1.0$, /bi/ $\delta = 1.0$. Top: Maeda parameters (Body orange). Middle: formants. Bottom: spectrograms — a single closure per word, no trailing schwa plateau.